THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A farmer grows wheat after pulses and applies less urea. The reason is: 1

(a) wheat needs no nitrogen **(b)** pulses consume nitrogen
(c) nitrogen left by pulses **(d)** urea is costly

2. A red rose in white light is viewed in a plane mirror. The image of the rose appears: 1

(a) red **(b)** green **(c)** white **(d)** black

3. A student claims 'the Sun is closer to Earth in summer'. A good test is: 1

(a) measure the temperature once
(b) look at the Sun
(c) ask friends
(d) compare dates of nearest distance with seasons

4. In a set-up with a plant in a bell jar with NaOH solution, starch is absent in the leaf because: 1

(a) NaOH adds oxygen **(b)** NaOH absorbs carbon dioxide
(c) light is blocked **(d)** NaOH gives CO2

5. The image formed by a plane mirror of a person 1.6 m tall is: 1

(a) 3.2 m tall **(b)** 0.8 m tall **(c)** 1.6 m tall **(d)** 0.4 m tall

6. If the Earth revolved around the Sun in half the time, a year would have about: 1

(a) 730 days **(b)** 182 days **(c)** 91 days **(d)** 365 days

7. Which of these is a net result of photosynthesis in a day? 1

(a) only oxygen **(b)** less glucose
(c) more glucose than used in respiration **(d)** no glucose

8. A student sees a shadow of her hand on a screen 2 m from a lamp. If she moves the hand to the midpoint between lamp and screen, the shadow size relative to the hand is: 1

(a) four times **(b)** same **(c)** half **(d)** double

9. The Sun is much larger than the Moon, yet in a total solar eclipse the Moon can hide it because the Moon is: 1

(a) larger **(b)** much nearer to the Earth
(c) brighter **(d)** much hotter

10. A student claims that mushrooms make food in sunlight. The correct reply is: 1

(a) mushrooms have no chlorophyll and absorb food from dead matter
(b) correct
(c) they use CO2
(d) they make starch

11. The colour of the shadow of a red ball in white light is: 1

(a) white **(b)** green **(c)** black **(d)** red

12. A student says that 'the Moon has phases because the Earth's shadow falls on it'. This is: 1

(a) correct for the first quarter **(b)** correct
(c) incorrect **(d)** partly correct

13. **Assertion (A):** Stomata help in the exchange of gases in a leaf. 1
Reason (R): Guard cells control the opening and closing of stomata.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The Moon is a non-luminous object. 1
Reason (R): The Sun is a luminous object.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The seasons are caused by the changing distance of the Earth from the Sun. 1
Reason (R): The axis of the Earth is tilted.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do the Sun and the Moon rise in the east and set in the west? 2

17. How do plants adapt to deserts to reduce transpiration? 2

18. Explain why the Moon looks bright at night but is not a luminous body. 2

19. Why do places at high latitudes have very long summer days? 2

20. Explain why the leaf in the dark does not show starch though it is green. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why does a wet black road at night show the lights of vehicles? 3

22. Discuss how the study of eclipses helped scientists, with two examples. 3

23. Explain the changes in a leaf's starch content from morning to night and the reason. 3

24. A girl is 4 m from a mirror and walks 1 m towards it. By how much does the distance between her and her image decrease? 3

OR

If the new moon occurs on 1 March, estimate the dates of first quarter, full moon and last quarter. 3

25. Explain with examples how plants help in keeping the air clean. 3

OR

Why are reflecting surfaces on roads and bicycles used? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Arun makes hand shadows of a bird on the wall of his room using a torch. When he moves his hand closer to the torch, the shadow becomes bigger. He also tries a sheet of clear glass and a sheet of butter paper in front of the torch. 5

(i) Write the three things needed to form a shadow.

(ii) What happens to the size of the shadow when the object is moved nearer to the torch?

(iii) What type of material is butter paper?

(iv) Will the clear glass sheet form a dark shadow? Give a reason.

27. A plant biologist finds a plant with no chlorophyll which grows on the bark of a tree. Discuss the possible modes of nutrition and how to decide. 5

OR

Design an experiment to find the relation between the object distance and the image distance in a plane mirror, with a table and conclusion. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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